$$\underbrace{D\tilde{\Theta} \xrightarrow[n\to\infty]{} 0 \quad \text{и } \tilde{\Theta} \text{ - несмещ}}_{\text{условие для выполнения условия достаточного условия состоятельности}}$$

$$D\tilde{\Theta} = D\left(\frac{2}{n}\sum_{i=1}^{n} x_i\right) = \frac{4}{n^2} D\left(\sum_{i=1}^{n} x_i\right) = \frac{4}{n^2}\sum_{i=1}^{n} Dx_i =$$

$$= \frac{4}{n^2}\cdot n\, Dx_i = \frac{\Theta^2}{12}\cdot\frac{4}{n} = \frac{\Theta^2}{3n} \xrightarrow[n\to+\infty]{} 0 \quad \forall \Theta > 0$$

$$Dx_i = \cancel{M\xi}\, D\xi = M\xi^2 - \underbrace{(M\xi)^2}_{= \frac{\Theta^2}{4}};$$

$$M\xi^2 = \int_0^\Theta x^2\cdot\frac{1}{\Theta}dx = \frac{x^3}{3\Theta}\Big|_0^\Theta = \frac{\Theta^2}{3}$$

$$\Rightarrow D\xi = \frac{4\Theta^2 - 3\Theta^2}{12} = \frac{\Theta^2}{12}$$

Итог: оценка несмещенная и состоятельная

б) $\tilde{\Theta}_2 = x_{min}$

• Несмещенность:

~~Из фо~~ Воспользуемся формулой из к-порядковой статистики:

$$\cancel{x_m}\ \tilde{\Theta}_2 \sim 1 - (1 - F(y))^n;$$

$$P = n(1 - F(y))^{n-1}\cdot(+P(y)) = n\left(1 - \frac{y}{\Theta}\right)^{n-1}\cdot\frac{1}{\Theta}\{(0,\Theta)\}$$

Тогда: $M\tilde{\Theta}_2 = \int_0^\Theta y\cdot n\left(1 - \frac{y}{\Theta}\right)^{n-1}\cdot\frac{1}{\Theta}dy =$

$$\cancel{= \frac{n}{\Theta}\int_0^\Theta y\left(1 - \frac{y}{\Theta}\right)dy = \frac{n}{\Theta}\frac{\Theta^2}{2} - \frac{y^3}{3\Theta}\Big|_0^\Theta\cdot\frac{n}{\Theta} = \frac{n}{2}\Theta - \frac{\Theta n}{3} = \frac{n}{6}\Theta}$$

$$= \left\{\begin{array}{l} 1 - \frac{y}{\Theta} = t;\ (1-t)\Theta = y \\ -dy = \Theta dt \end{array}\right\} = -\frac{n}{\cancel{\Theta}}\int_1^0 (1-t)\cancel{\Theta}\, t^{n-1}\cdot\Theta dt =$$

$$= n\Theta\int_0^1 (t^n - t^{n-1})dt = n\Theta\int_0^1 (t^{n-1} - t^n)dt = n\Theta\left(\frac{t^n}{n} - \frac{t^{n+1}}{n+1}\right)\Big|_0^1 =$$

$$= n\Theta\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{n\Theta(\cancel{n}+1-\cancel{n})}{\cancel{n}(n+1)} = \frac{\Theta}{n+1}$$

<u>~~Смещённая~~</u>

• Тогда несмещенная оценка: $\tilde{\Theta}_2' = x_{min}\cdot(n+1)$, т.к

$$M\tilde{\Theta}_2' = \Theta$$

• ~~Нес~~ Состоятельность:

$$D\tilde{\Theta}_2' = D(x_{min}(n+1)) = (n+1)^2 Dx_{min} = (n+1)^2\cdot\left(Mx_{min}^2 - \underbrace{(Mx)^2}_{= \Theta^2/(n+1)^2}\right)$$